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CSCE 190: Computing in the Modern World

Class 9 – Problem Solving with Computing – (GenAI)

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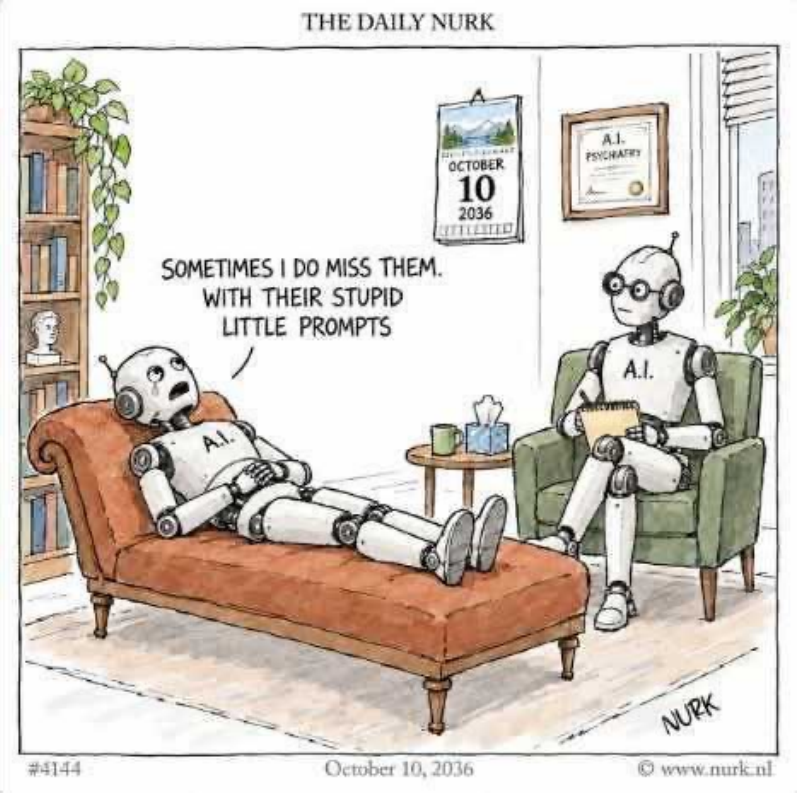
OCT 2026

Carolinian Creed: "I will practice personal and academic integrity."

Credits: **Copyrights of all material reused acknowledged**

Outline

- Introduction Section
  - Recap from Last Class
- Main Section
  - Problem Solving with
    - Computing (Software, Hardware, Services)
    - GenAI
  - Putting the Problem Solving Process Together
- Conclusion Section
  - Ask Me Anything



Introduction Section

Recap from Last Lecture - Class 8

We talked about

- Problem Solving Process
- **Validating the Problem**
  - **Creating Requirements and Specifications**
  - **Testing – Concepts and Activities**
- Selecting Problems that Matter

Assignment 8: Same for All in a Team

- Group 4-5 students
- Discuss a problem category
- Decide on a problem – see ***problems that matter***
- Write a problem statement
  - The group will have one testing plan
  - Add your statement to your GitHub in a sub-folder called 'assignment-8'
    - Do mention all the students in your group

- Work with your group
  - If you have a group, sit with them
  - If you don't, join an existing group.
- Discuss a system setup
  - Inputs and outputs
  - Metrics
- Discuss **evaluation**
  - Functional specification
  - Non-functional specification
- Write test plans
  - A test cases has a sample inputs and expected outputs
  - Q: How many testcases to create ?

See sample testcases at: https://github.com/biplav-s/course-cmw-f26/tree/main/software-dev

Computing in the News

1. **AI system submits false homicide tip to Philadelphia police,**
https://www.washingtonpost.com/technology/2026/10/09/ai-system-submits-false-homicide-tip-philadelphia-police/, Oct 2026

- The tip was submitted in July through the Philadelphia Police Department's online tip line for unsolved murders and "purported to come from someone who might have information about the case," the department said in a statement in Oct 2026. The tip was flagged by an automated system and never acted upon. Anthropic discovered the incident on Sept. 28 (two months late) and notified Philadelphia Police.
- Filing false police reports is illegal in Pennsylvania.

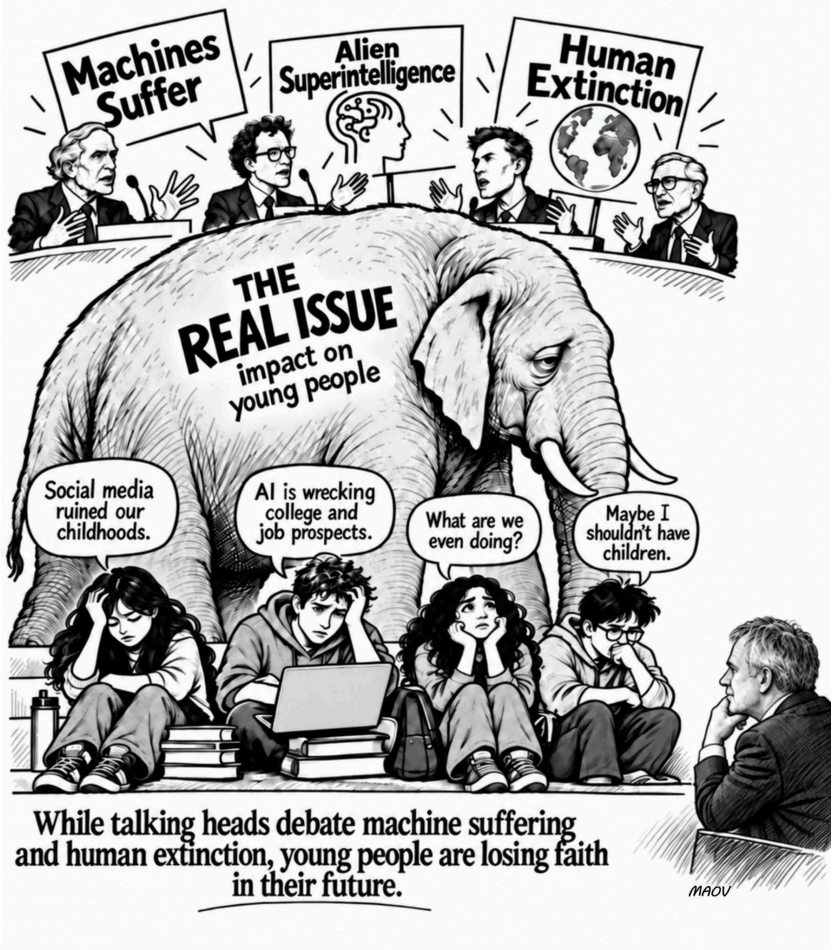
2. **Companies may be held liable when AI harms their clients and customers,**
https://theconversation.com/companies-may-be-held-liable-when-ai-harms-their-clients-and-customers-282234, Oct 2026

- The National Institute of Standards and Technology issued a report on AI risk management in July 2024 that calls AI mistakes and hallucinations "**confabulations**" – the confident presentation of erroneous or false information
- A 2024 Stanford University study of specialized AI tools for legal research found erroneous results in 17% to 33% of the responses tested.

Oct 2026

Safe AI needed for
- Youth
- Seniors

An event: https://tinyurl.com/safe-ai-2026



Main Section

Problem Solving Process

Receiving the problem ***{from client}***

- Client creates **requirements** document

- **Understanding the problem and scoping**
  - Developer creates **specification**



- **Solving**
  - Understand and implementing with tools for solving the problem: **Design docs, test plans**
  - Create solutions: **Code, Release**
  - Evaluate solutions: **Test logs**

- **Delivering *{to customers}***

A generic process involves:

- Customer:
  - requirements
- Developer:
  - specification
  - **design**
  - **development (coding)**
  - **testing**
  - **release**
- Customer:
  - evaluation
- acceptance

The above process is highly simplified and rarely happens in one-round. Read about 'Software Development Life Cycles' (SDLCs).

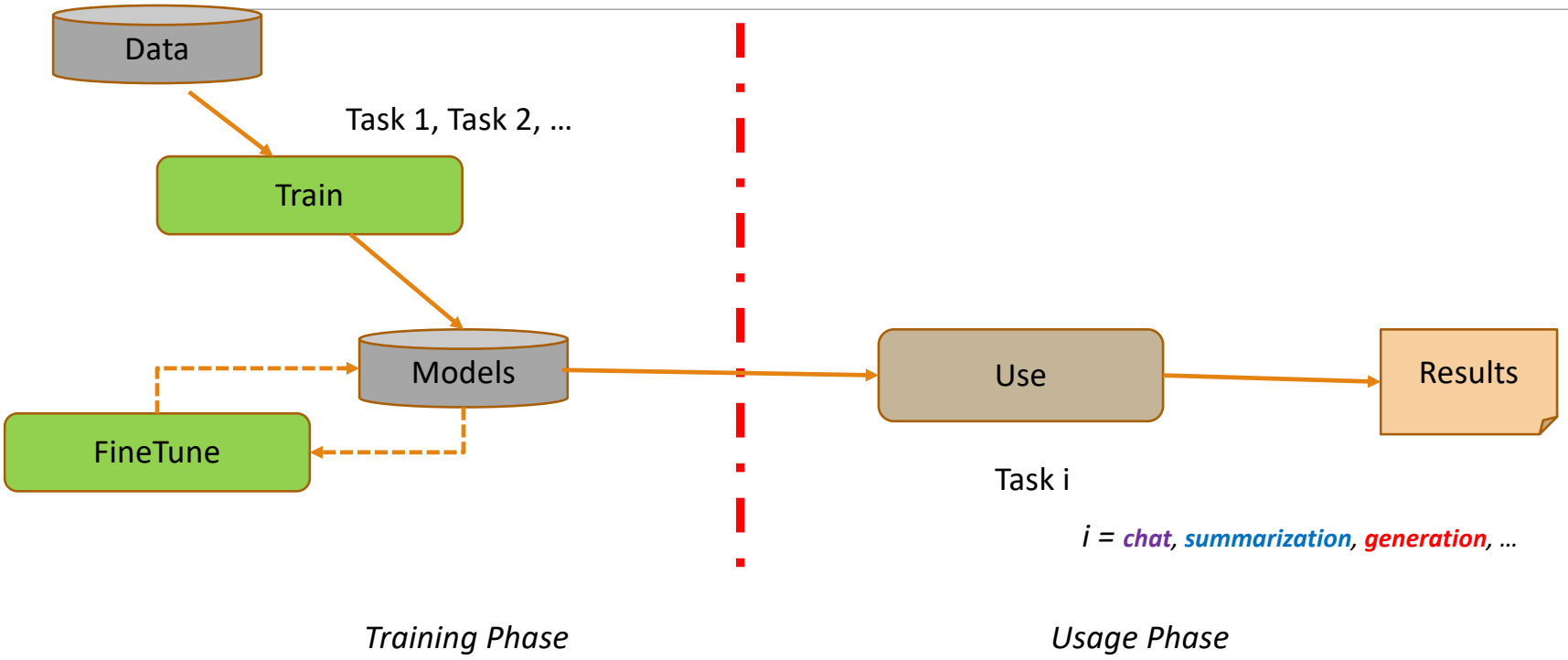
Solving with Computing

- Software-only – e.g., writing a program
- Hardware-only – e.g., installing and operating device
- Services-only – e.g., helping the customer giving advice directly
- Solutions
  - Any combination of software, hardware and services
  - E.g., "helping approaching pedestrians about uneven sidewalks" via **a dynamic sign-board**

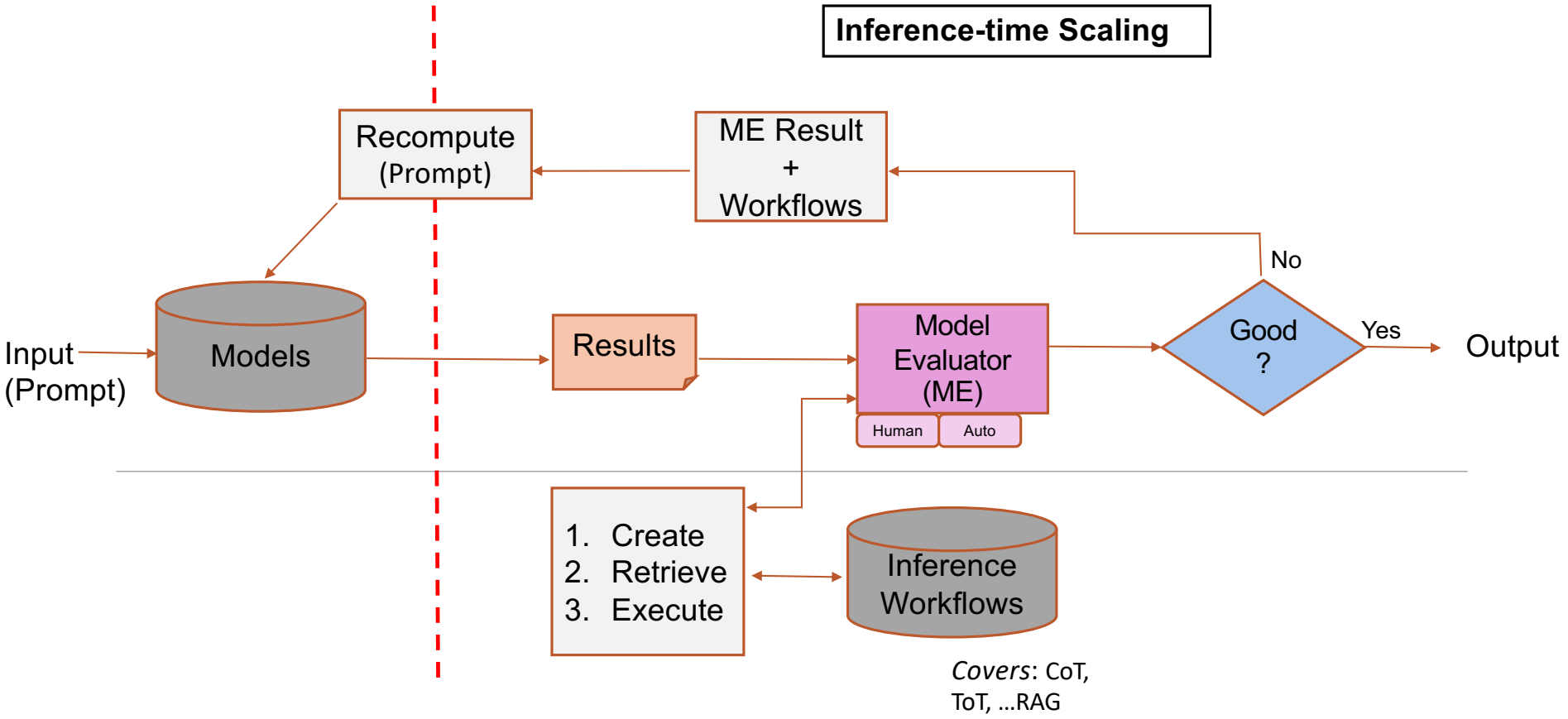
Solving with Generative AI

- **Pros**
  - Getting a quick and “personal” informational response
  - Convenient due to low barrier for use in many cases
- **Cons**
  - Response may be erroneous
  - Responses may be hard to verify, especially if the users in ignorant about the area of inquiry
  - Data may be leaked to GenAI provider
  - Usage may be costly without knowing why
    - Immediately due to token usage.
    - Long-term due to adverse impact environment per usage
    - **Relationship between inquiry, response and taken use is not transparent**

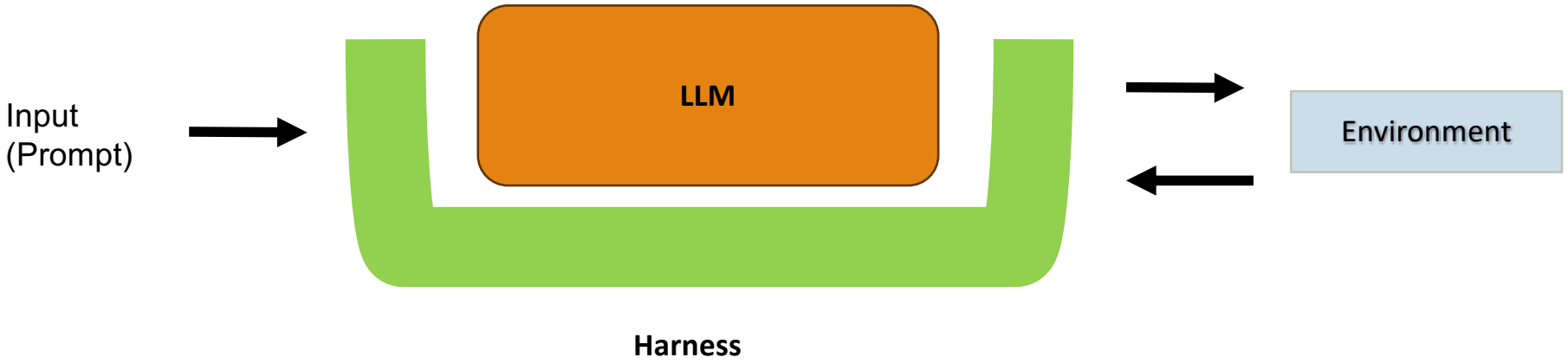
Large Language Models (LLMs) Basics



Inference Time with LLMs



Building an *AI Agent**

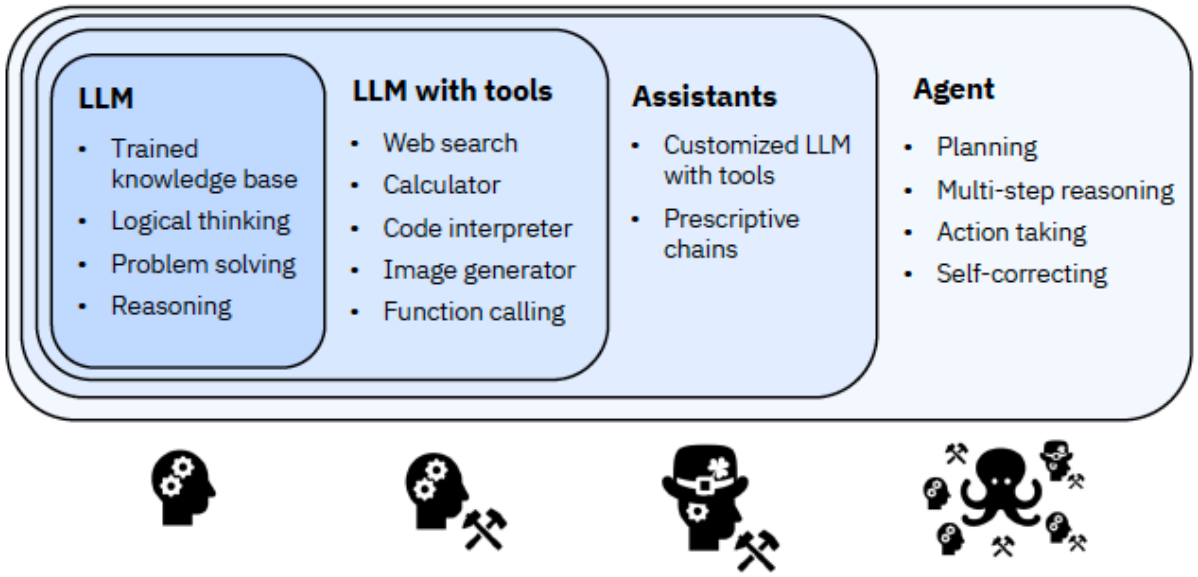


*An **agent harness**, also known as **agent scaffolding**, is the software infrastructure surrounding a large language model (LLM) that enables it to operate as an AI agent.*

***Examples**: Anthropic's Claude Code, OpenAI's Codex, Google DeepMind's Antigravity, SpaceXAI's Cursor, Qwen, DeepSeek, Kimi, and GLM.*

Source:
https://en.wikipedia.org/wiki/Agent_harness

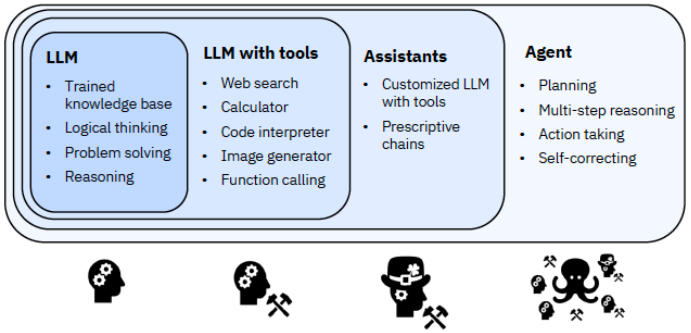
LLM + Extensions

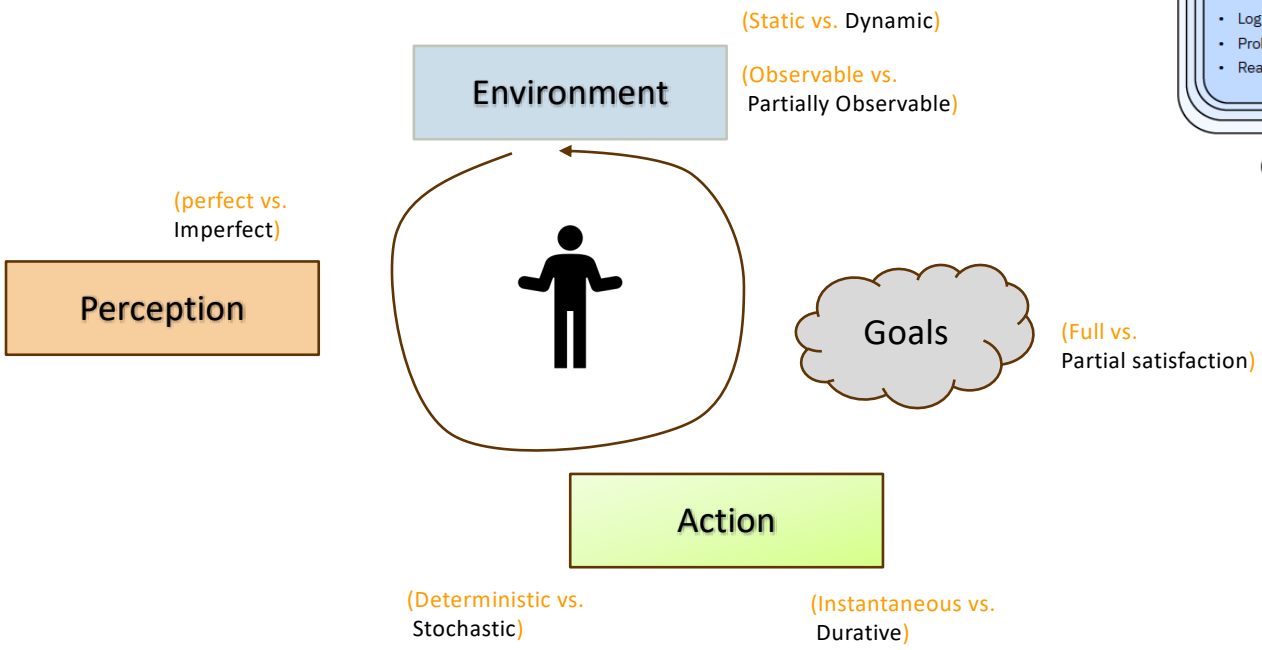


Credit: K. Varshney's talk - **Individual and Collective Human Agency in the Face of 'AI'**, 2026

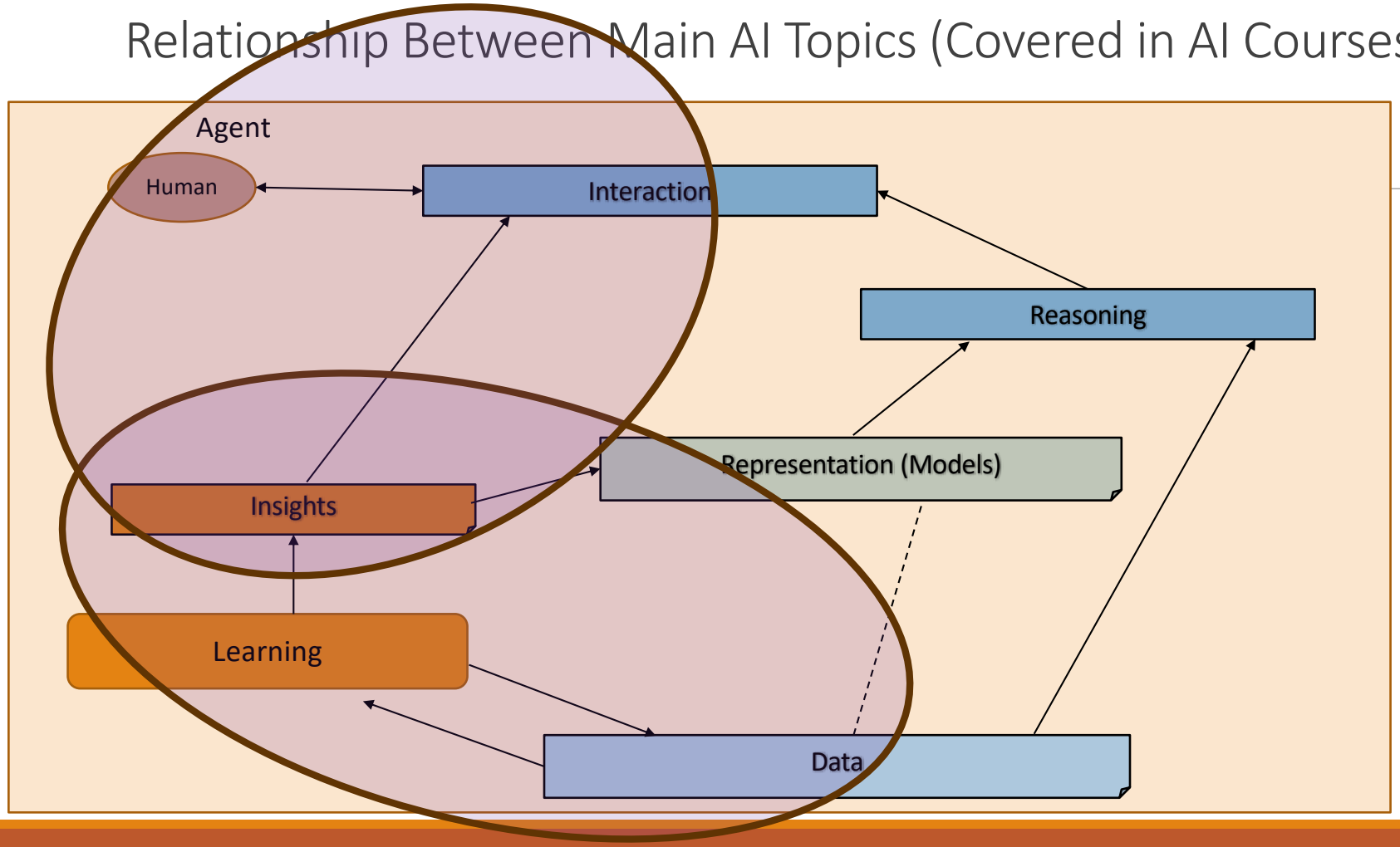
Intelligent Agent Model vis-a-vis Agentic AI

Credit: K. Varshney's talk - **Individual and Collective Human Agency in the Face of 'AI'**, 2026

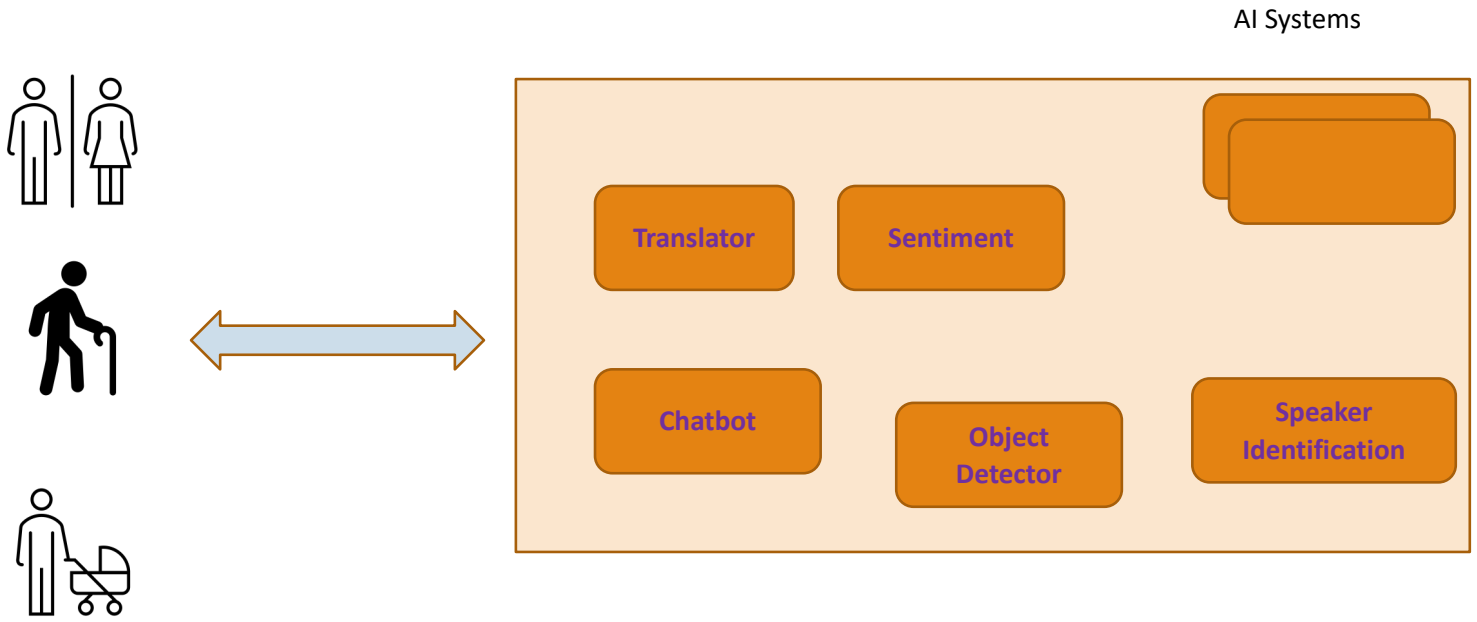




Relationship Between Main AI Topics (Covered in AI Courses)

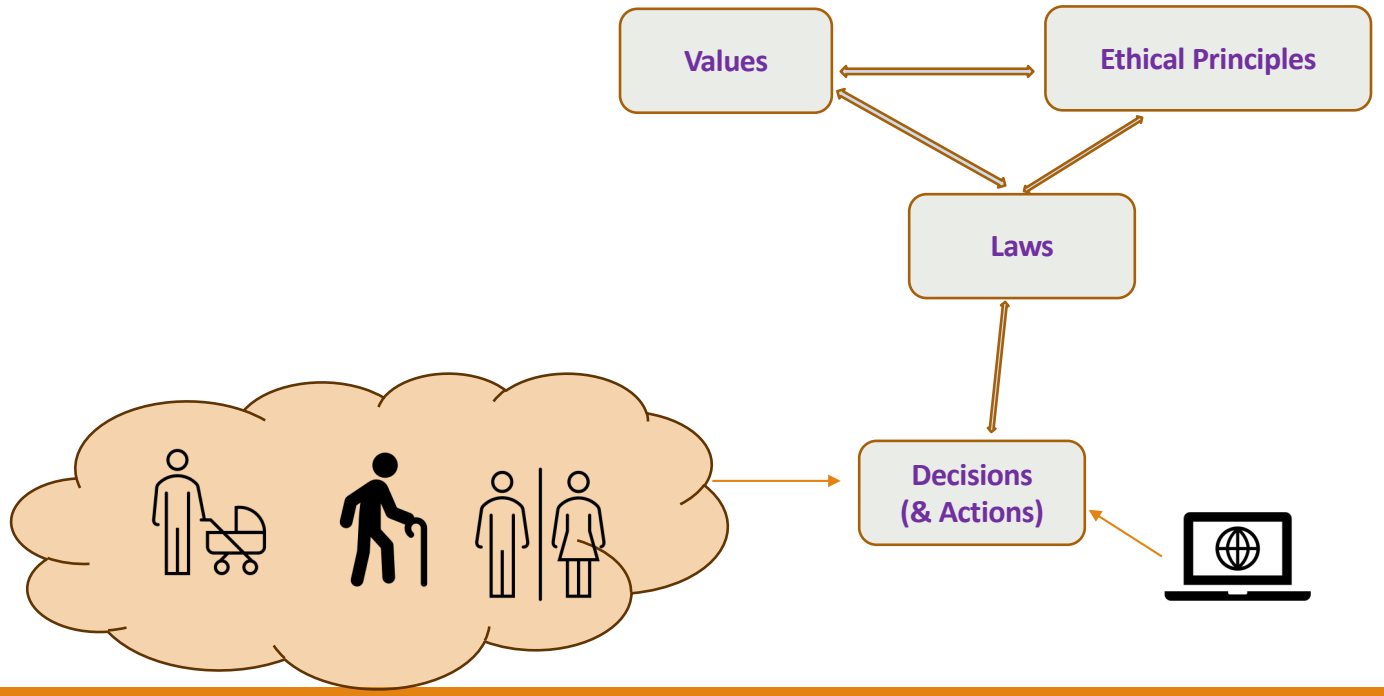


Technology and People



Trust: *Can people trust* ***new system*** *to perform capably, consistently, and with human values?*

Values, Ethics, Law, Decisions (Actions) with Trust



Trust: *how can,*
humans or machines (AI) or in collaboration together,
*take decisions and actions following the same **values**, **ethical** principles and **laws ?***
Assumption: *they should follow same values, principles and laws.*

Problem Example 1: Student Meals Reco

- **Problem**: build a system to help students choose meals during study days
- Validating the problem:
  - The inputs: e.g. data, current process
  - The outputs (deliverables): e.g., software (app)
  - Criteria for problem being solved: e.g., adoption rate
- Solving the problem
  - Recommendation methods
- Trustworthiness ecosystem
  - Values, ethical principles, laws
  - Users impacted

Problem statement criteria

- Are focused on the user (avoid using “we” or “the product”)
- Are broad (leave room for innovation and alternatives)
- Are manageable (just this one problem)

Discussion: Using GenAI for Example 1

Problem Example 2: Nobel Prizes

- **Problem**: build a system to contextualize about Nobel Prizes wins
- Validating the problem:
  - The inputs: e.g. data, current process
  - The outputs (deliverables): e.g., software (app)
  - Criteria for problem being solved: e.g., adoption rate
- Solving the problem
  - **Identifying topics and trends**
- Trustworthiness ecosystem
  - Values, ethical principles, laws -
  - Users impacted – ***students making future decisions based on trends ?***

Problem statement criteria

- Are focused on the user (avoid using "we" or "the product")
- Are broad (leave room for innovation and alternatives)
- Are manageable (just this one problem)

Example: TestCases – Nobel Prize Info System

- Functional types
  - Given info for valid
    - Prize categories
    - Years
    - Persons
- Non-function types
  - Performance related
  - Handling of sensitive categories related

Nobel Prizes: Non-functional Requirements/ Specification

- Winners background
  - Name, gender, age, country, institution
  - Should or shouldn't be given ? Any disclaimer
- What do the awards in a year tell about
  - The past ?
  - The future ?
- Themes
  - Terminology changes – how to handle them

Discussion: Using GenAI for Example 2

Snapshot from Different Sections

Section 1	Section 2	Section 4
1. students on scooters (harm) 2. Students managing money in college 3. Students finding housing (off-campus) 4. Students managing assignments **5. (all) Sidewalks uneven** 6. Students managing finances – 1st year 7. Students: saving meal swipes (7!)	**1. Students: textbook overprices and requirements** 2. (All) drunk driving at 5-points **3. Students: on-campus laundry** 4. Students: USC bus trackers are unreliable 5. Students: non-reserved parking 6. Students: double-deck parking	1. Students: pedestrian safety, accessibility 2. Students: campus parking and transportation 3. {} – problem not noted 4. Students: college tuition payment **5. Students: time management and resources**

Assignment 9: Same for All in a Team

- Group 4-5 students
- Discuss a problem category
- Decide on a problem – see ***problems that matter***
- Solve for your problem
  - The group will have one solution plan
  - Add your statement to your GitHub in a sub-folder called 'assignment-9'
    - Do mention all the students in your group

- Create prompt(s)
  - Record response(s) and iterate, if necessary
- Discuss prompts and responses
  - Verify (with specification, test plans)
  - Validate (with requirements, test plans)
  - Consider both functional specification and non-functional aspects
- Re-think solving strategy
  - LLM choice(s)
  - Prompt strategies
  - Solving approach

See sample testcases at: https://github.com/biplav-s/course-cmw-f26/tree/main/software-dev

Concluding Section

Conclusion

We talked about

- Problem Solving Process
- Validating the Problem
- Selecting Problems that Matter

Week	Content	Assignment
1	Intro, syllabus, university resources	Slack account
2	CSE majors, career paths, research areas	Write a mission statement
3	Github (account and basics) + Lab	Github account, repo
4	HTML basics, IDE (VS Code)	None
5	More HTML, CSS, Problem Solving	Update Github with CSS
6	Brainstorming, problem statement	Problem statement
7	Computing – (Software) Dev. Life Cycle	Draft Requirement, Specification
8	Computing – Testing a System	Test Plan
9	AI – LLMs/FMs, Generative AI, Issues and Challenges	Problem Solving with GenAI
10	Building a resume, LinkedIn	Draft resume in PDF
11	Review resumes	Review resumes, update Github
12	Convert resume to HTML	Begin conversion, update Github
13	CSE research areas	(Finalize website)
N/A	Last day of classes: 4 Dec, 2026	**All extensions end**

Ask Me Anything